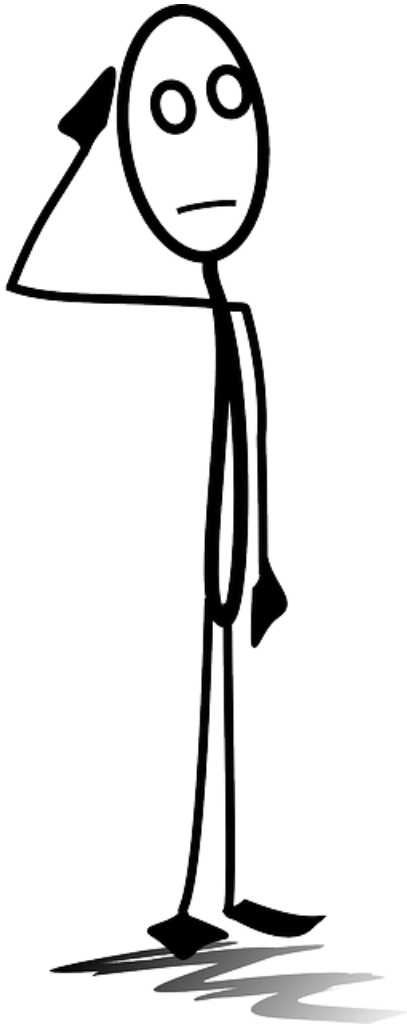
PHYSICS I
WEEK 5

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Week 5: outline

- Work-Energy Theorem: Taylor 4.1, 4.2
- Stable, and unstable equilibrium: Taylor 4.6
- Introduction to phase space: Thornton & Marion 3.4



WORK ENERGY THEOREM

Kinetic Energy

$$T = \frac{1}{2}mv^2$$

WORK ENERGY THEOREM

Kinetic Energy

$$T = \tfrac{1}{2}mv^2 \quad \Rightarrow \quad \frac{dT}{dt} = \tfrac{1}{2}m\frac{d}{dt}(\mathbf{v}\cdot\mathbf{v})$$

WORK ENERGY THEOREM

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$$T = \tfrac{1}{2}mv^2 \quad \Rightarrow \quad \frac{dT}{dt} = \tfrac{1}{2}m\frac{d}{dt}(\mathbf{v}\cdot\mathbf{v}) = \tfrac{1}{2}m(\dot{\mathbf{v}}\cdot\mathbf{v} + \mathbf{v}\cdot\dot{\mathbf{v}}) = m\dot{\mathbf{v}}\cdot\mathbf{v}$$

$$\frac{dT}{dt} = \mathbf{F}\cdot\mathbf{v}$$

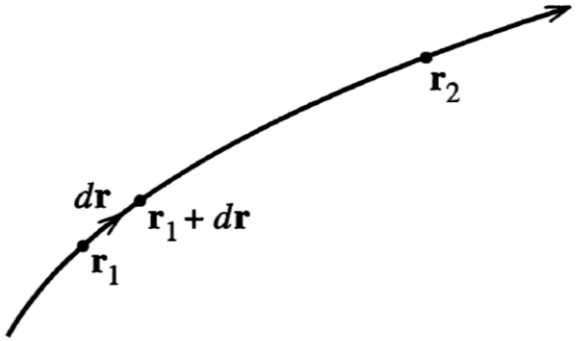
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$$\frac{dT}{dt} = \mathbf{F}\cdot\mathbf{v} \quad \Rightarrow \quad dT = \mathbf{F}\cdot d\mathbf{r}$$

$$\Delta T \equiv T_2 - T_1 = \sum \mathbf{F}\cdot d\mathbf{r}$$

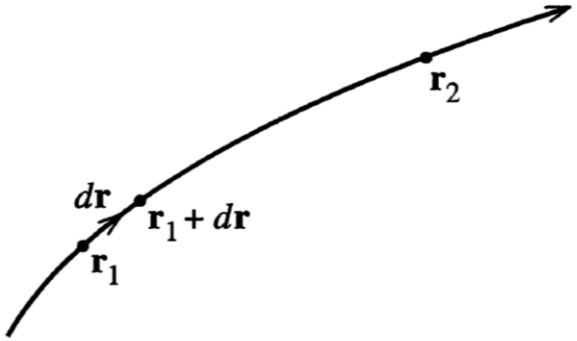


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$$\Delta T \equiv T_2 - T_1 = \sum \mathbf{F}\cdot d\mathbf{r}$$

In the limit that all the displacements $d\mathbf{r}$ go to zero

$$\Delta T \equiv T_2 - T_1 = \int_1^2 \mathbf{F}\cdot d\mathbf{r} \qquad \equiv W(1 \to 2)$$

The line integral

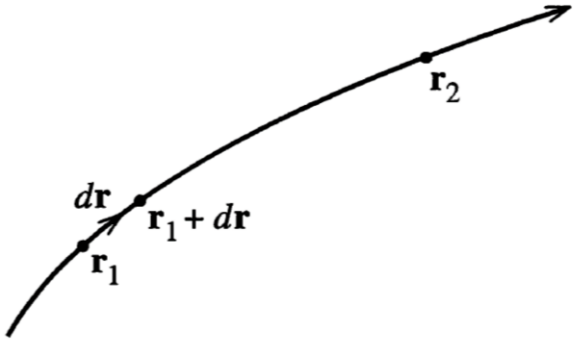
Work done *by the force moving between points 1 and 2*

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Work done *by the force moving between points 1 and 2*

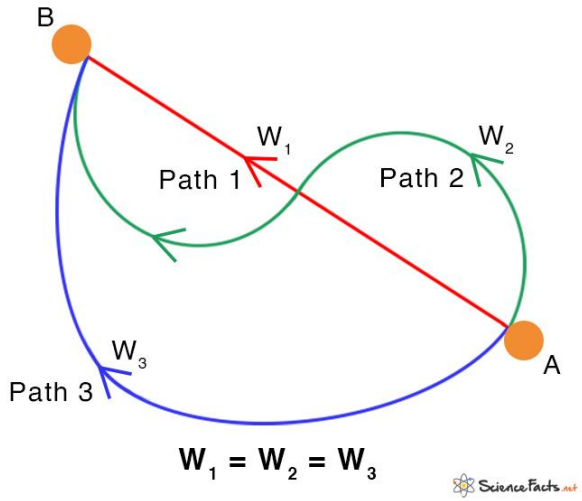
Remember – ***F*** *is the net force!*

$$\mathbf{F} = \mathbf{F}_1 + \cdots + \mathbf{F}_n \equiv \sum_{i=1}^{n} \mathbf{F}_i$$

$$T_2 - T_1 = \sum_{i=1}^{n} W_i(1\rightarrow 2)$$

WORK ENERGY THEOREM

Conservative Force



Conditions for a Force to be Conservative

A force **F** acting on a particle is **conservative** if and only if it satisfies two conditions:

(i) **F** depends only on the particle's position **r** (and not on the velocity **v**, or the time t, or any other variable); that is, $\mathbf{F} = \mathbf{F}(\mathbf{r})$.

(ii) For any two points 1 and 2, the work $W(1 \rightarrow 2)$ done by **F** is the same for all paths between 1 and 2.

POTENTIAL ENERGY

$$\Delta T \equiv W(1 \to 2)$$

$$U(\mathbf{r}) = -W(\mathbf{r}_o \to \mathbf{r}) \equiv -\int_{\mathbf{r}_o}^{\mathbf{r}} \mathbf{F}(\mathbf{r}') \cdot d\mathbf{r}'$$

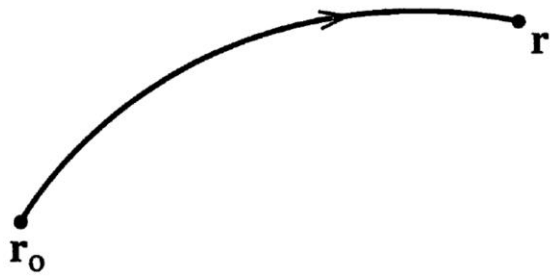


Figure 4.4 The potential energy $U(\mathbf{r})$ at any point $\mathbf{r}$ is defined as minus the work done by $\mathbf{F}$ if the particle moves from the reference point $\mathbf{r}_o$ to $\mathbf{r}$. This gives a well-defined function $U(\mathbf{r})$ only if this work is independent of the path followed — that is, the force is conservative.

Mechanical Energy

$$E = \text{KE} + \text{PE} = T + U(\mathbf{r})$$

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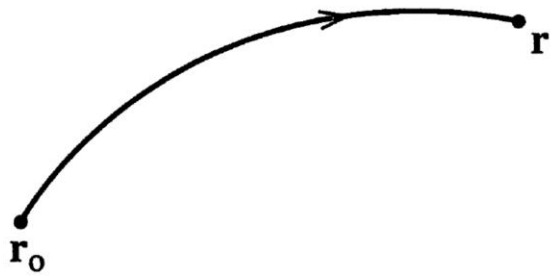


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Mechanical Energy

$$E = \text{KE} + \text{PE} = T + U(\mathbf{r})$$

$$W(\mathbf{r}_1 \to \mathbf{r}_2) = W(\mathbf{r}_o \to \mathbf{r}_2) - W(\mathbf{r}_o \to \mathbf{r}_1)$$

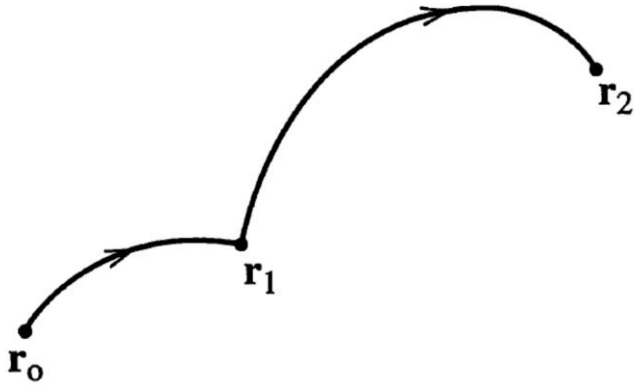


Figure 4.5 The work $W(\mathbf{r}_1 \to \mathbf{r}_2)$ going from $\mathbf{r}_1$ to $\mathbf{r}_2$ is the same as $W(\mathbf{r}_o \to \mathbf{r}_2)$ minus $W(\mathbf{r}_o \to \mathbf{r}_1)$. This result is independent of what path we use for either limb of the journey, provided the force concerned is conservative.

$$W(\mathbf{r}_1 \to \mathbf{r}_2) = -[U(\mathbf{r}_2) - U(\mathbf{r}_1)] = -\Delta U$$

$$\Delta T = -\Delta U$$

CONSERVATION OF ENERGY

In the presence of several conservative forces:

Principle of Conservation of Energy for One Particle

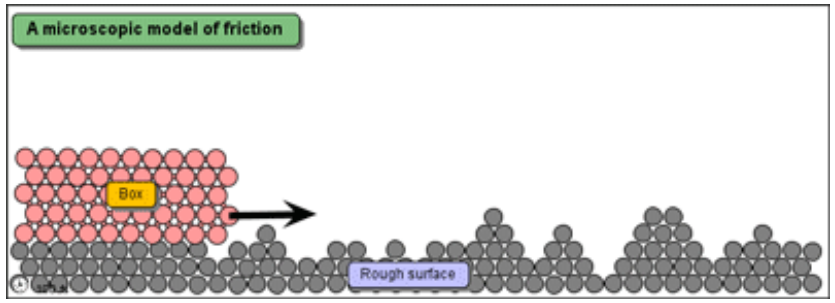
If all of the n forces $\mathbf{F}_i$ $(i = 1, \cdots, n)$ acting on a particle are conservative, each with its corresponding potential energy $U_i(\mathbf{r})$, the **total mechanical energy**, defined as

$$E \equiv T + U \equiv T + U_1(\mathbf{r}) + \cdots + U_n(\mathbf{r}), \tag{4.22}$$

is constant in time.

CONSERVATION OF ENERGY

In the presence of non-conservative force:



$$\Delta T = W = W_{\text{cons}} + W_{\text{nc}}$$

$$\Delta E \equiv \Delta(T + U) = W_{\text{nc}}$$

EXAMPLE: LINE INTEGRALS

Evaluate the line integral for the work done by the two-dimensional force $\mathbf{F} = (y, 2x)$ going from the origin O to the point $P = (1, 1)$ along each of the three paths shown in Figure 4.2. Path a goes from O to $Q = (1, 0)$ along the x axis and then from Q straight up to P, path b goes straight from O to P along the line $y = x$, and path c goes round a quarter circle centered on Q.

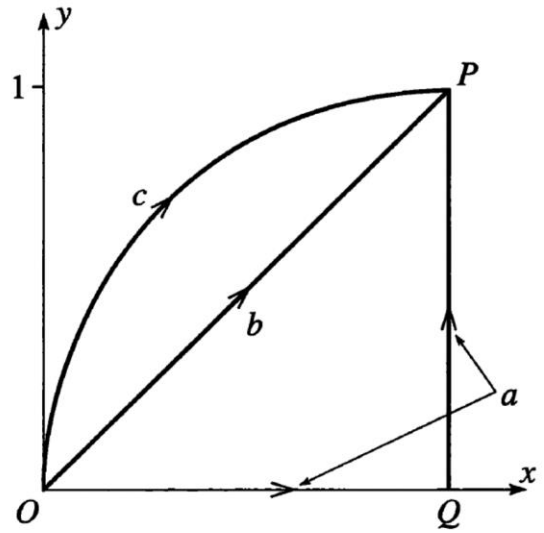
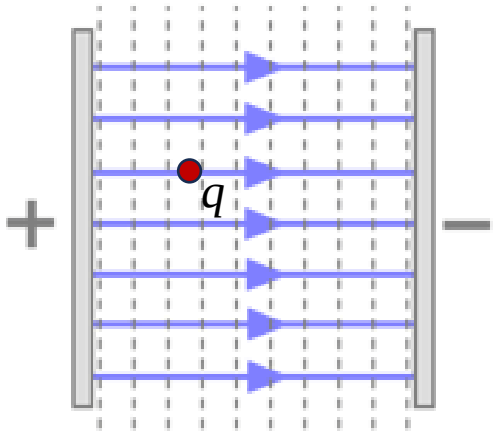


Figure 4.2 Three different paths, a, b, and c, from the origin to the point $P = (1, 1)$.

EXAMPLE: CONSERVATIVE FORCE

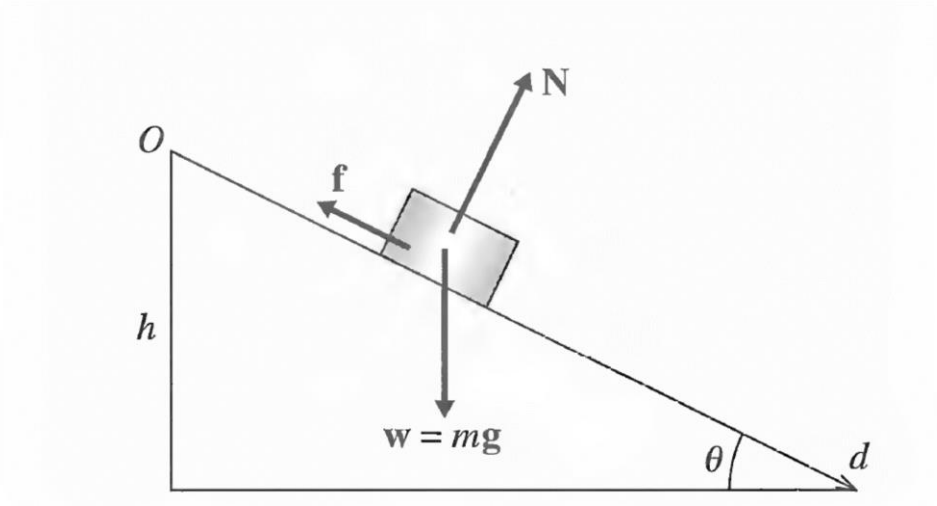
A charge q is placed in a uniform electric field pointing in the x direction with strength E_0, so that the force on q is $\mathbf{F} = q\mathbf{E} = qE_0\hat{\mathbf{x}}$. Show that this force is conservative and find the corresponding potential energy.



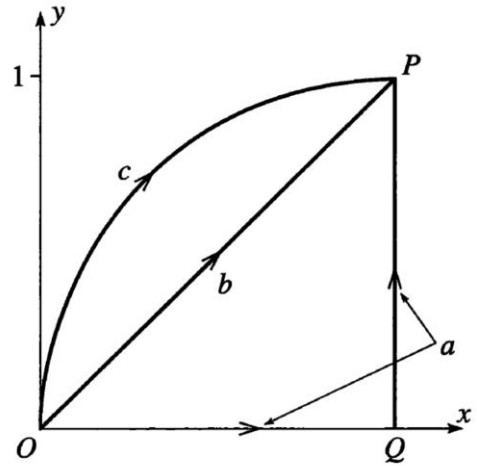
EXAMPLE: NON-CONSERVATIVE FORCE

A block of mass m is observed accelerating from rest down an incline that has coefficient of friction μ and is at angle θ from the horizontal.

find its speed v when it reaches the bottom of the slope, a distance d from its starting point.



REVIEW: EXAMPLES

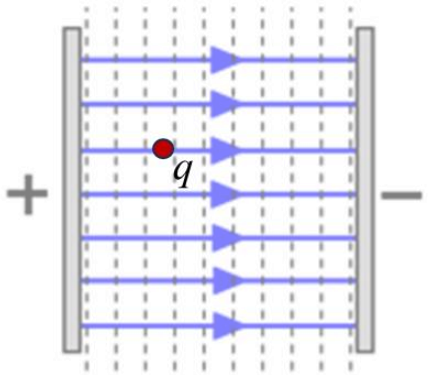


$$\mathbf{F} = (y, 2x)$$

$$W_a = \int_a \mathbf{F}\cdot d\mathbf{r} = \int_O^Q \mathbf{F}\cdot d\mathbf{r} + \int_Q^P \mathbf{F}\cdot d\mathbf{r}$$

$$= \int_0^1 F_x(x,0)\,dx + \int_0^1 F_y(1,y)\,dy$$

$$= 0 + 2\int_0^1 dy = 2$$

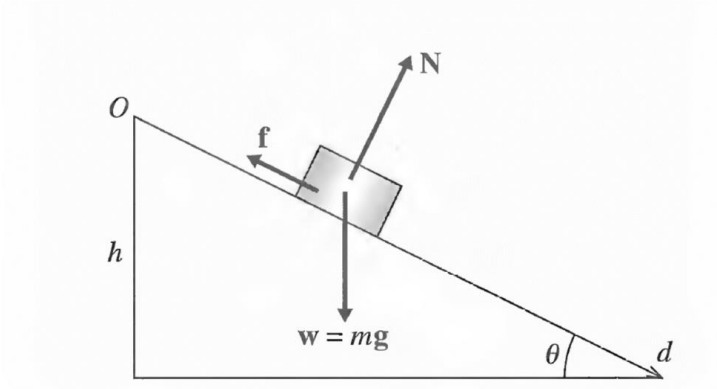


$$\mathbf{F} = q\mathbf{E} = qE_o\hat{\mathbf{x}}$$

$$W(1\to 2) = \int_1^2 \mathbf{F}\cdot d\mathbf{r} = qE_o\int_1^2 \hat{\mathbf{x}}\cdot d\mathbf{r}$$

$$= qE_o\int_1^2 dx = qE_o(x_2 - x_1)$$

$$U(\mathbf{r}) = -qE_o x$$



$$\Delta T + \Delta U = W_{\text{fric}}$$

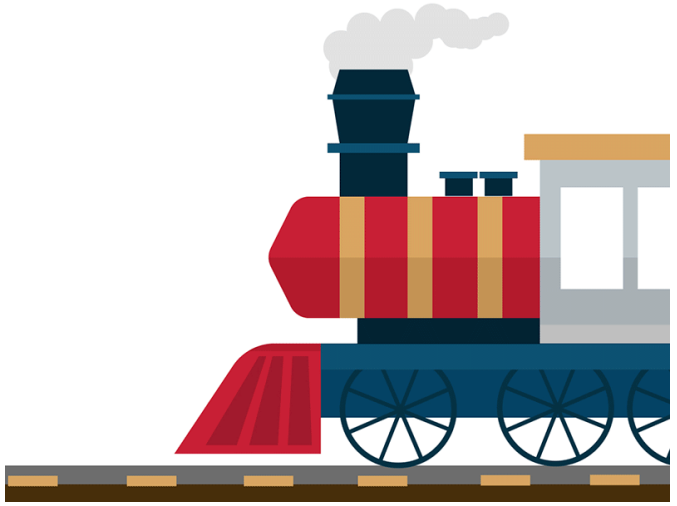
$$\tfrac{1}{2}mv^2 - mgd\sin\theta = -\mu mgd\cos\theta$$

$$v = \sqrt{2gd(\sin\theta - \mu\cos\theta)}$$

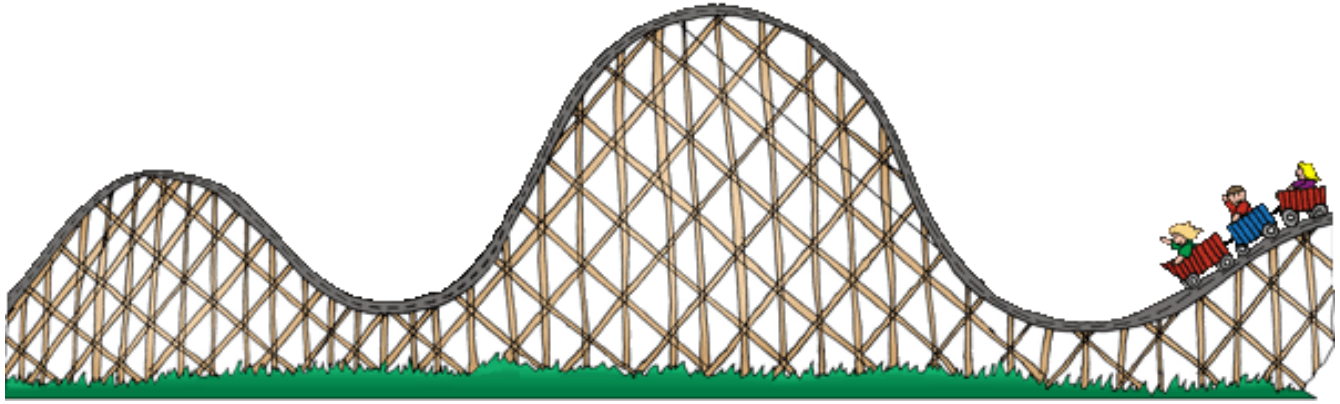
LINEAR 1D SYSTEMS



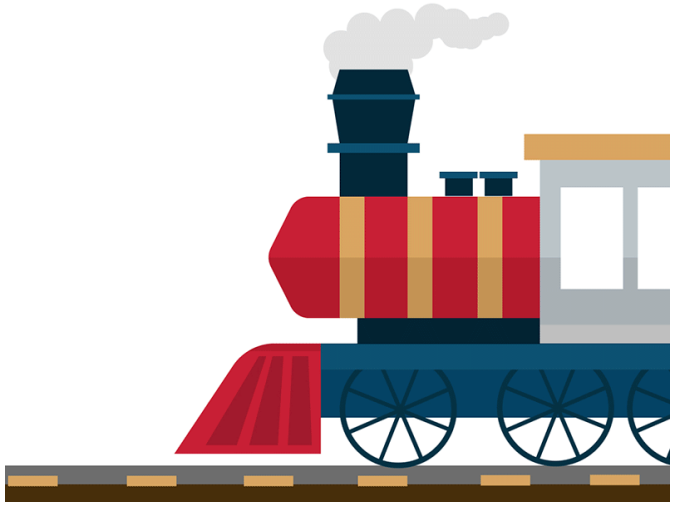
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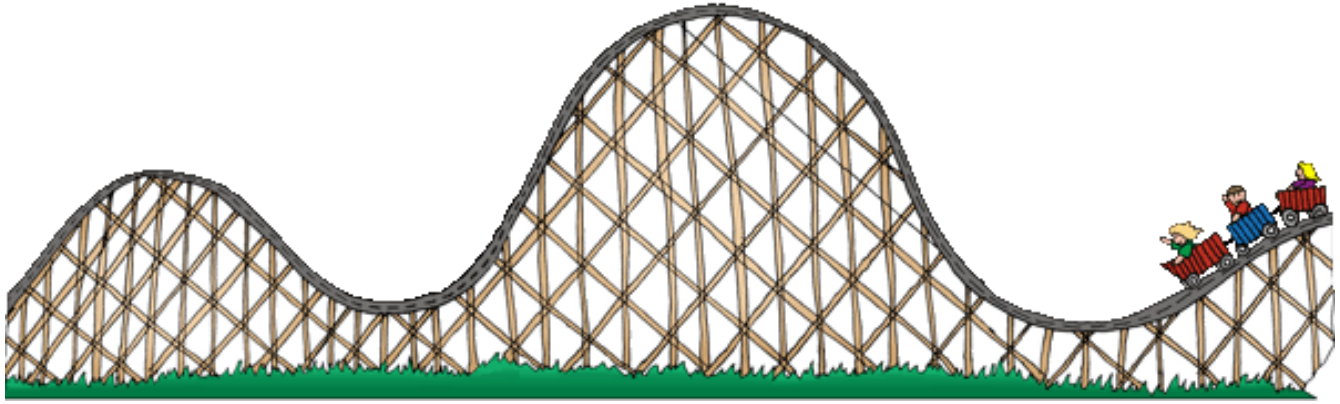
$$W(\boldsymbol{r_1} \rightarrow \boldsymbol{r_2}) = \int_{\boldsymbol{r_1}}^{\boldsymbol{r_2}} \boldsymbol{F}(\boldsymbol{r}).\boldsymbol{dr} \quad \Rightarrow \quad W(x_1 \rightarrow x_2) = \int_{x_1}^{x_2} F_x(x)\,dx$$



LINEAR 1D SYSTEMS



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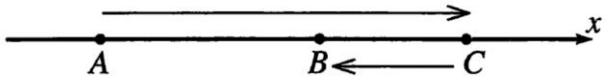


Figure 4.9 The path called $ABCB$ goes from A past B and on to C, then back to B.

$$W(ABCB) = W(AB)$$

LINEAR 1D SYSTEMS: POTENTIAL ENERGY

$$U(x) = -\int_{x_0}^{x} F_x(x')\,dx' \quad \Rightarrow \quad F_x = -\frac{dU}{dx}$$

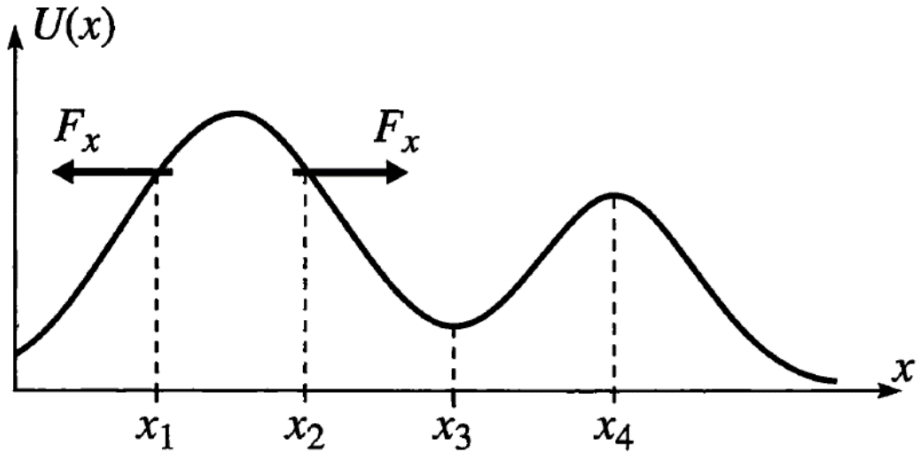
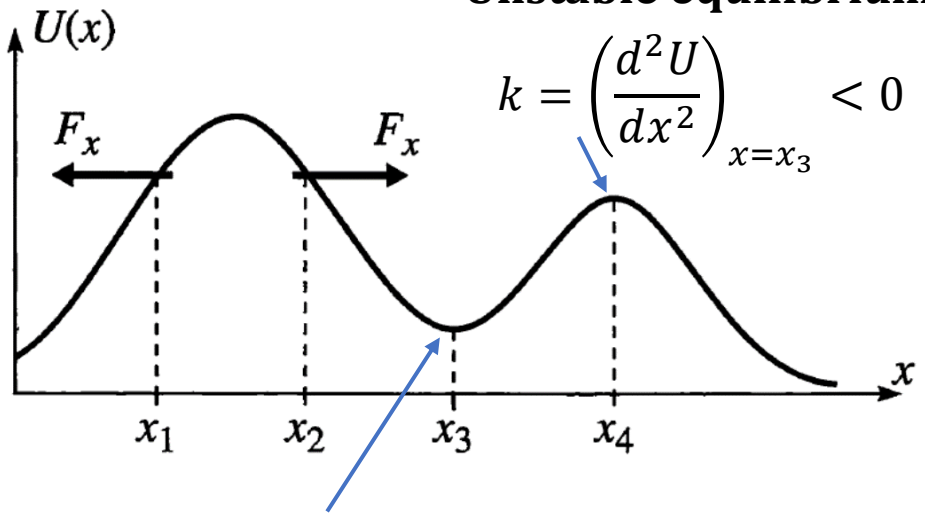


Figure 4.10 The graph of potential energy $U(x)$ against x for any one-dimensional system can be thought of as a picture of a roller coaster track. The force $F_x = -dU/dx$ tends to push the object "downhill" as at x_1 and x_2. At the points x_3 and x_4, where $U(x)$ is minimum or maximum, $dU/dx = 0$ and the force is zero; such points are therefore points of equilibrium.

LINEAR 1D SYSTEMS: POTENTIAL ENERGY

$$U(x) = -\int_{x_0}^{x} F_x(x')\,dx' \quad \Rightarrow \quad F_x = -\frac{dU}{dx}$$

Unstable equilibrium



$$k = \left(\frac{d^2U}{dx^2}\right)_{x=x_3} < 0$$

Stable equilibrium (due to the restoring force)

$$k = \left(\frac{d^2U}{dx^2}\right)_{x=x_3} > 0 \qquad \omega = \sqrt{\frac{k}{m}}$$

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LINEAR 1D SYSTEMS: PHASE SPACE

The conservation of energy is actually sufficient to "solve" the problem of one-dimensional motion in the following sense. The phase space in this case has just two dimensions (it is the "**phase plane**"). The equation

$$\frac{1}{2}m\,\dot{x}^2 + U(x) = E \tag{5.3}$$

represents a curve in the $(x,\ \dot{x})$ plane, which is nothing but the phase trajectory. Of course, to find $x(t)$ explicitly as a function of t, we do have to solve the equation of motion.

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$$\dot{x}(x) = \pm\sqrt{\frac{2}{m}}\sqrt{E - U(x)}$$

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$$\dot{x}(x) = \pm\sqrt{\frac{2}{m}}\sqrt{E - U(x)} \quad \Rightarrow \quad t = \int_{x_0}^{x} \frac{dx'}{\dot{x}(x')} = \sqrt{\frac{m}{2}}\int_{x_0}^{x} \frac{dx'}{\sqrt{E - U(x')}}$$

LINEAR 1D SYSTEMS: EXAMPLE

Linear harmonic oscillator $U(x) = Kx^2/2 \text{ with } K > 0$

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For any $E > 0$

$x_- \leq x \leq x_+$, where $x_\pm = \pm\sqrt{2E/K}$. ***Turning points***

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Each phase trajectory is an ellipse given by

$$\frac{1}{2}m\dot{x}^2 + U(x) = E \quad \Rightarrow \quad \boxed{\frac{x^2}{(2E/K)} + \frac{\dot{x}^2}{(2E/m)} = 1}$$

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First-order differential equation

LINEAR 1D SYSTEMS: EXAMPLE

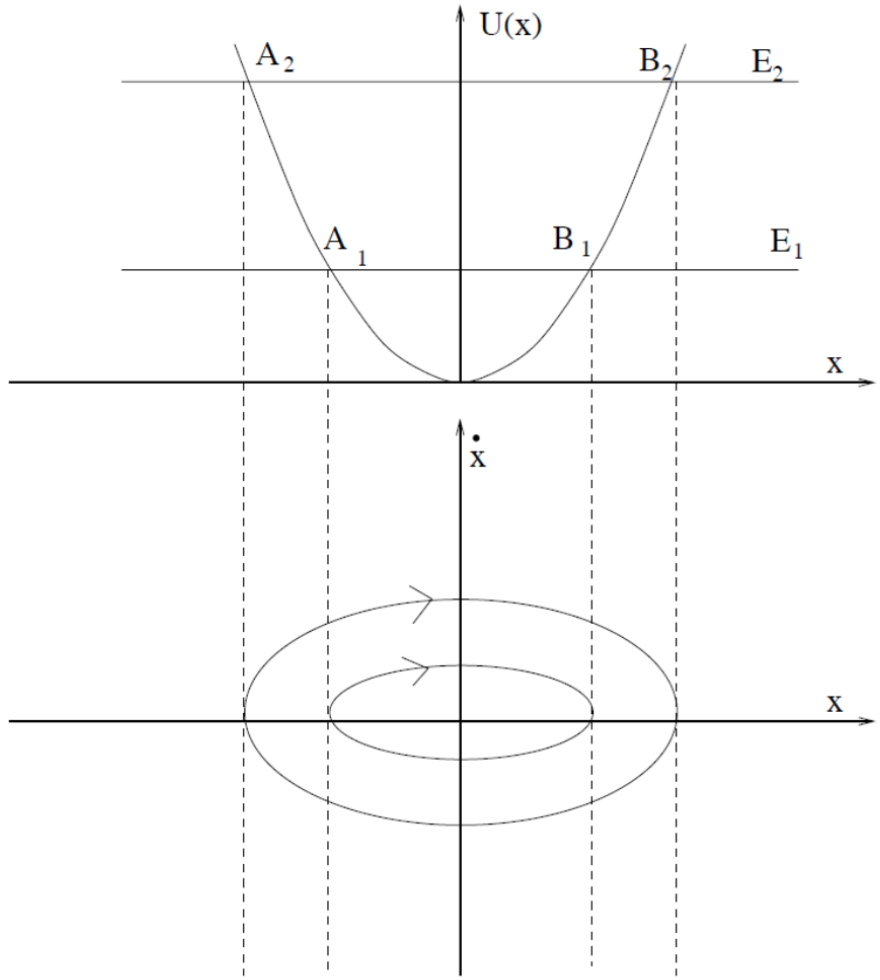
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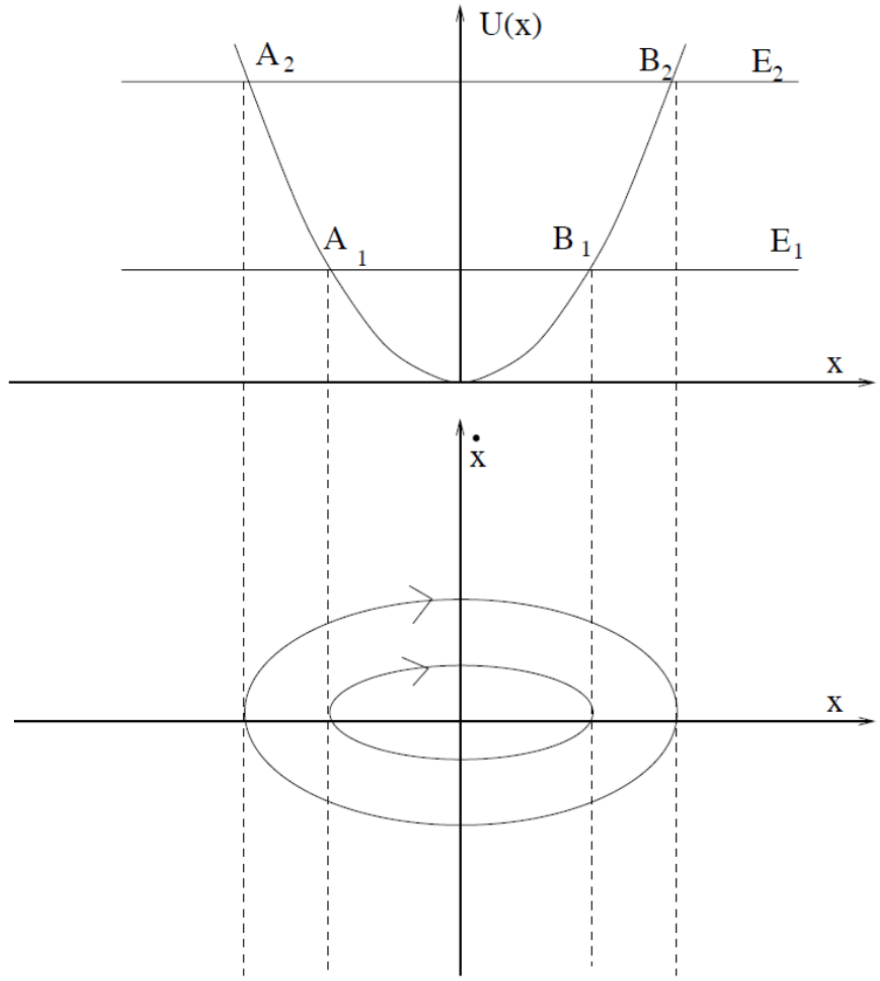
- Every E > 0 specifies a particular ellipse. The phase portrait is a family of such ellipses in the phase plane.
- The initial conditions determine the particular ellipse on which the system lies, as well as the precise point on it at which the motion starts.
- The sense or direction in which the ellipse is traversed as time progresses is not arbitrary – it is clockwise, as indicated by the arrows.

LINEAR 1D SYSTEMS: EXAMPLE

Linear harmonic oscillator $U(x) = Kx^2/2$ with $K > 0$

Each phase trajectory is an ellipse given by

$$\frac{1}{2}m\dot{x}^2 + U(x) = E \quad \Rightarrow \quad \frac{x^2}{(2E/K)} + \frac{\dot{x}^2}{(2E/m)} = 1$$

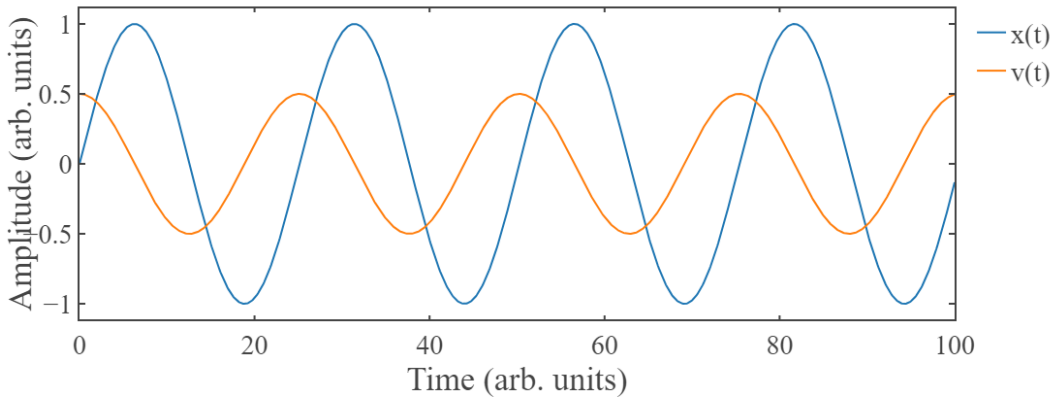


$$\ddot{x} = -\frac{k}{m}x = -\omega^2 x$$

Initial conditions

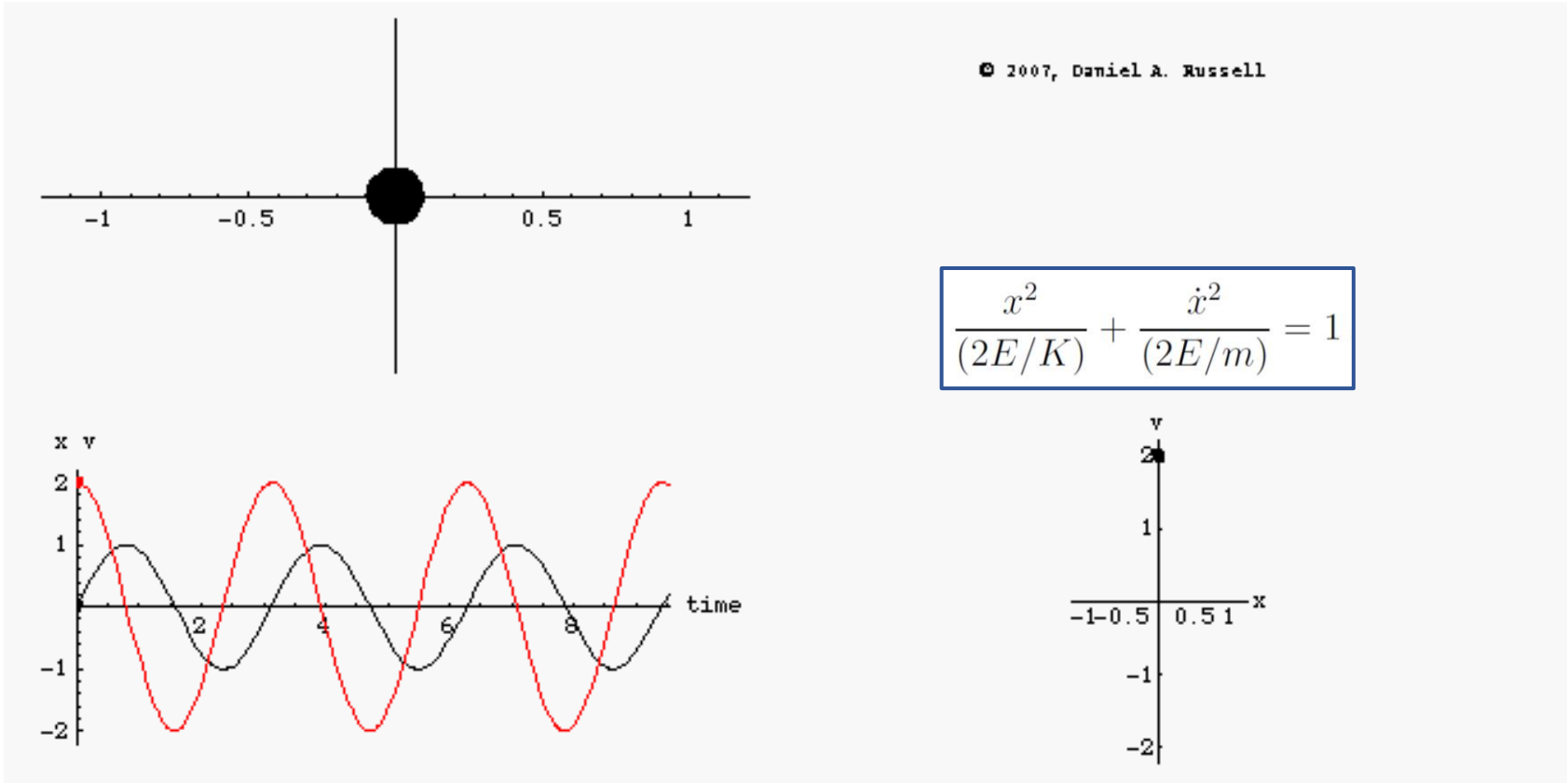
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$$x(t) = \frac{v_0}{\omega}\sin(\omega t)$$



LINEAR 1D SYSTEMS: EXAMPLE

Linear harmonic oscillator $U(x) = Kx^2/2$ with $K > 0$



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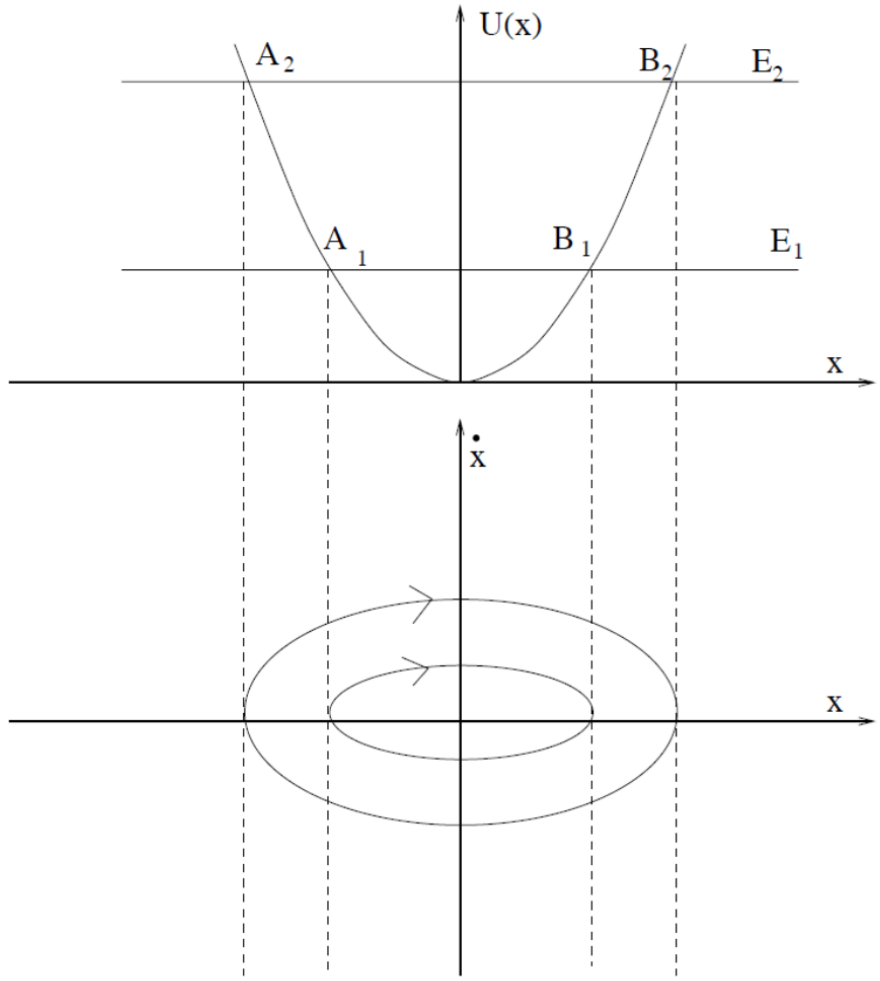
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$$T = 2\int_{A_1}^{B_1} \frac{\sqrt{m}dx}{\sqrt{(2E - Kx^2)}} \quad \Rightarrow \quad T = 2\pi(m/K)^{1/2}$$

